LE NGUYEN GIA HUNG

AI Undergraduate Researcher | Time-Series & Multimodal XAI

Research interests: explainable AI, time-series forecasting, multimodal learning and computer vision.

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EDUCATION

FPT University — Ho Chi Minh City, Vietnam

B.Eng. in Artificial Intelligence; GPA: 8.62/10.0 — *2023 – Dec 2027 (expected)*

- Certificate of Merit: Honorable Student of Trimester, semesters 3, 4, 5 and 6.

PUBLICATIONS & MANUSCRIPTS

Peer-reviewed conference publication

Gia-Hung Nguyen Le, Gia-Bao Pham Hoang, Tan-Phat Vo, Thu Le and Nhu Nguyen. "Proactive Air Quality Forecasting and Health Alert System for Melbourne." *AJCAI 2025, Springer LNAI*. First and corresponding author. DOI: 10.1007/978-981-95-4969-6_34.

Manuscripts under review

"Stress-Testing Multi-Source Cyanobacterial Bloom Forecasting under Sparse Monitoring." *Under review, AJCAI 2026*. First and corresponding author.

"Warning Before the Evening Peak: Health-Centred Air Quality Alerts in New Zealand and Australia." *Under review, AJCAI 2026*. Second author.

"Trustworthy Multimodal Anomaly Detection in Time Series with Conformal Calibration and Counterfactual Attribution." *Under review, ACM SIGKDD 2027*. Second author.

SELECTED RESEARCH EXPERIENCE

SmokeNet: Air Quality Forecasting — Academic research

First & Corresponding Author; published at AJCAI 2025 — *Jun – Nov 2025*

- Designed the Transformer forecasting architecture with seasonal embeddings and reconstruction loss for Melbourne's $PM_{2.5}$.
- Reported MAE **0.7470**, R^2 **0.9545** and **35.9% lower MAE than XGBoost** on the 2024 held-out evaluation.
- Evaluated the May 2024 smoke event separately: **57.7% lower MAE than the baseline** in that backtest.
- Built a chronological evaluation pipeline on 4+ years of OpenWeather data; used SHAP, Integrated Gradients and DiCE to inspect forecasts.

Cyanobacterial Bloom Forecasting — Academic research

First & Corresponding Author; AJCAI 2026 under review — *Jan 2026 – Present*

- Built a causal weekly benchmark across **7 New Zealand lakes**: $n = 5{,}432$, with an untouched 2023–2024 holdout ($n = 170$).
- Compared forecast horizons: 7-day Chl-a history XGBoost reached weighted F1 **0.791** (+0.103 over full fusion, $p = 0.016$); 14-day full fusion reached **0.703**.
- Led manuscript writing and the 8-gate provenance audit: 22 apparent episode starts reduced to 3 eligible incidents.

Health-Centred Air Quality Alerts (NZ & Australia) — Academic research

Second Author; AJCAI 2026 under review — *Jan 2026 – Present*

- Evaluated exposure coverage across Canterbury's 10-year ($n = 620{,}057$) and Melbourne's 4-year ($n = 823{,}009$) hourly records.
- Reported that daily-mean triggers overstate exposure protection by 7.7–18.6 points; 42% of exposure occurred on sub-threshold daily-mean days.
- Built an archived weather-forecast panel model: 1-day-ahead MAE **3.19** $\mu g/m^3$, R^2 **0.54**.

Trustworthy Multimodal Anomaly Detection Academic research
Second Author; ACM SIGKDD 2027 under review *Jan 2026 – Present*

- Audited look-ahead text leakage across **9 domains**: a reported 4.52% error reduction from naive text fusion reversed under embargo (leakage fraction 1.50).
- Developed C-DLE with split-conformal calibration and Mondrian pools for false-alarm control under missing modalities.
- Designed cross-modal counterfactual attribution; reported **94.3% accuracy** against ground-truth anomalies with a **10% coreset**.

MERR-GAT: Multimodal Emotion Recognition Academic research
Second & Corresponding Author; in progress *Oct 2025 – Present*

- Develop **WavLM + Compact Conformer** audio encoding and **BERT/RoBERTa** text representations, fused through dual-channel graph attention.
- Inspect modality contributions using attention visualization, SHAP and Integrated Gradients; attention alone is not a causal explanation.
- Evaluation is in progress on **IEMOCAP, RAVDESS and MELD**, using WA, UA and Macro-F1.

Vietnamese Speech Recognition Academic research
Co-Researcher & Co-Author; in progress *Sep 2025 – Present*

- Build a **5.4M-parameter** ASR model: 4-layer CNN, 3-layer residual BiLSTM and CTC training on **745 hours** of Vietnamese speech.
- Reported **WER 33.09%** and **CER 15.28%**; compare with PhoWhisper to study the efficiency–accuracy trade-off. Method and evaluation notes.

PPE Detection (YOLOv8) Computer vision
Developer *Course project*

- Built 9-class detection, safe/unsafe compliance logic and a Streamlit webcam interface with violation reporting.

LEADERSHIP & CLOUD LEARNING

SpeedyLabX Student Research Group FPT University
Founder & Student Lead *Jan 2025 – Present*

- Lead a 10-member undergraduate group, with weekly technical workshops, paper reading and shared conference-submission workflows; research spans 3 pilots and 4 manuscripts.

First Cloud AI Journey Program AWS Vietnam program
FCAJ Cloud Intern, Team NeuraX *2026*

- Documented cloud learning, workshops and project work in the program journal.

SKILLS

Programming: Python, SQL/PostgreSQL; TypeScript, React, FastAPI.
ML & evaluation: PyTorch, Transformers, TensorFlow/Keras, Scikit-learn, OpenCV; SHAP, Integrated Gradients, DiCE, chronological cross-validation.
Data & tools: Pandas, NumPy, Matplotlib, Weights & Biases, Jupyter, Git, Docker, Linux, Streamlit. Technical writing: manuscripts, documentation and blog posts.

CERTIFICATIONS

DeepLearning.AI: Natural Language Processing.
Coursera: Gradient to Production: MLOps & Model Serving.
IBM: Full Stack Software Developer; Data Science Fundamentals with Python and SQL; Statistics for Data Science with Python; Introduction to Containers with Docker, Kubernetes & OpenShift.
Packt: AI Engineer Professional.